

File + Solns.

Student Number: _____



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Oxley College

TRIAL EXAMINATION 2000

3 UNIT MATHEMATICS

Time Allowed - 2 hours
(Plus 5 minutes reading time)

INSTRUCTIONS

- * Attempt **all** questions.
- * All questions are of **equal** value.
- * Show all necessary working in every question.
- * Marks may be deducted for poorly arranged or careless work.
- * *Board-approved* calculators may be used.
- * Clearly label each question and part on your answer sheet.
- * Start each question on a new page.

Question 1.**(12 marks)****(Start on a new page)**

- [4] (a) (i) Write down the expansion of $\tan(A + B)$
(ii) Find the value of $\tan 105^\circ$ in simplest surd form.

[3] (b) Solve the inequality: $\frac{2x + 1}{2x - 1} \geq 2$

[2] (c) Evaluate: $\int_0^{\pi/4} \cos x \sin^2 x \, dx$.

[3] (d) Solve: $x^6 - 9x^3 + 8 = 0$

Question 2.**(12 marks)****(Start on a new page)**

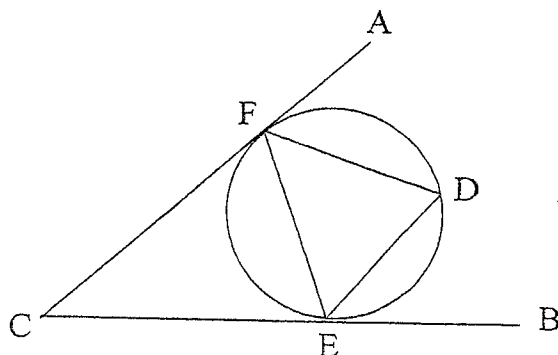
[4] (a) (i) Find $\frac{d}{dx} (x \log x - x)$

(ii) Hence evaluate $\int_2^e \log x \, dx$. Leave the answer in exact form.

- [3] (b) In the diagram, AC and BC are tangents to the circle, touching at F and E respectively. $\angle ACB$ equals 50° .

(i) Show that $\angle CEF$ is 65° .

(ii) Hence, find $\angle EDF$, giving reasons for your answer.



5] (c) (i) Show that $\cos 6x = 2\cos^2 3x - 1$.

(ii) The arc of the curve $y = \cos 3x$ between the lines $x = 0$ and $x = \frac{\pi}{6}$ is rotated about the x -axis.

Using (i), or otherwise, find the exact volume of the solid formed.

Question 3.**(12 marks)****(Start on a new page)**

- [5] (a) Consider the equation $x^2 - 4 + \log_e x = 0$
- (i) Show, by means of calculations, that the root of the equation lies between 1 and 2.
 - (ii) Use two applications of the 'halving the interval' method to find a smaller interval containing the root.
 - (iii) By drawing a graph of $y = \log_e x$, and any other appropriate graph on the same set of axes, verify that the equation has only one root.
- [7] (b) Given the function: $y = \frac{x^2 - 2x - 3}{x - 1}$
- (i) Find the coordinates of the points of intersection with the axes.
 - (ii) Find the equation of any asymptotes.
 - (iii) Show that the curve has no stationary points.
 - (iv) Sketch the curve.

Question 4.

(12 marks)

(Start on a new page)

- [4] (a) The roots, α , β and γ of the equation $8x^3 - 36x^2 + 22x + 21 = 0$ are in an arithmetic progression.

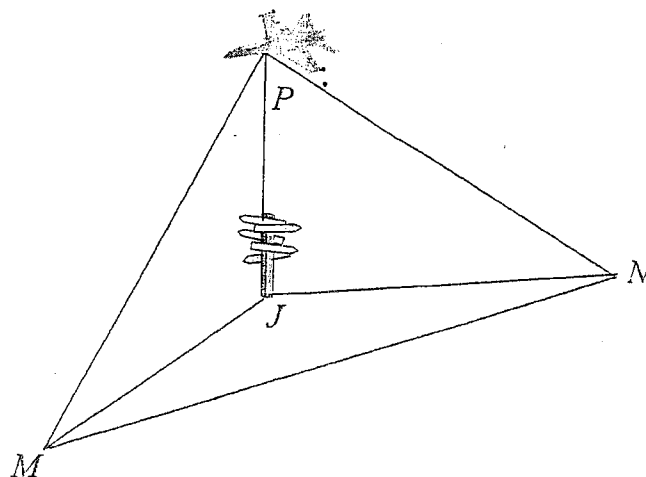
- (i) Show that $\alpha + \gamma = 2\beta$
- (ii) Write down the value of $\alpha + \beta + \gamma$
- (iii) Find α , β and γ .

- [4] (b) (i) Show that $\sum_{r=1}^n (5r-4) = 1 + 6 + 11 + \dots + (5n-4)$

- (ii) Hence, prove by Mathematical Induction that $\sum_{r=1}^n (5r-4) = \frac{1}{2}n(5n-3)$

- 4] (c) From a plane 500 metres above a road junction J , the angle of depression to a point M , due south of the junction is 42° . To another point N , bearing 080° from the junction, the angle of depression is 32° .

- (i) Find the lengths of JM and JN
- (ii) How far apart are M and N ?

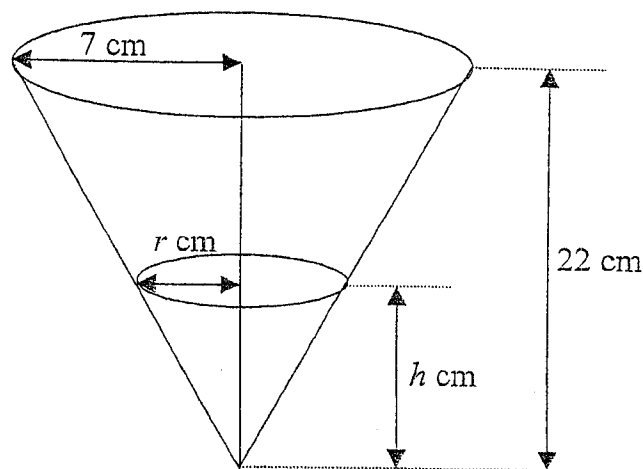


Question 5.**(12 marks)****(Start on a new page)**

- [7] (a) $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ are two points on the parabola $x^2 = 4ay$. The variable chord PQ is such that it is always parallel to the line $y = x$.
- (i) Find the gradient of PQ and hence show that $p + q = 2$
 - (ii) Show that the equation of the **normal** at P is: $x + py = 2ap + ap^3$, given that the gradient of the *tangent* at P equals p .
 - (iii) Write down the equation of the normal at Q , and hence find the coordinates of the point of intersection R , of these normals.
 - (iv) Prove that the locus of R is the straight line $x - 2y + 12a = 0$
- [5] (b) Newton's Law of Cooling states that when an object at temperature $T^\circ\text{C}$ is placed in an environment at temperature $T_0^\circ\text{C}$, the rate of the temperature loss is given by the equation:
- $$\frac{dT}{dt} = k(T - T_0) \quad \text{where } t \text{ is the time in seconds and } k \text{ is a constant}$$
- (i) Show that $T = T_0 + Ae^{kt}$ is a solution to the equation.
 - (ii) A packet of peas, initially at 24°C is placed in a snap-freeze refrigerator in which the internal temperature is maintained at -40°C . After 5 seconds, the temperature of the packet is 19°C .
 - (α) Show that the value of the constant A is 64.
 - (β) Show that the value of the constant $k \approx -0.0163$
 - (γ) How long will it take for the packet's temperature to reduce to 0°C ?

Question 6.**(12 marks)****(Start on a new page)**

- [2] (a) Find the general solution for: $\tan 2x = 1$
- [5] (b) A golf ball is lying on a horizontal fairway when a golfer hits it. It just passes over a 2.25 metre high tree 1.5 seconds later. The tree is 60 metres away from the point from which the ball was hit. Taking $g = 10 \text{ m s}^{-1}$, calculate:
- (i) the initial velocity and the angular projection.
 - (ii) how far away, from where the golfer hits it, does the ball land.
- [5] (c) Soft serve ice cream is served into a jumbo-sized right circular cone as shown in the diagram. The cone has height 22 cm and radius 7 cm.



Ice cream is leaking through a hole of negligible size in the bottom of the cone at a constant rate of 7 cm^3 per minute.

- (i) Use similar triangles to find a relationship between r and h .
- (ii) Show that when the depth of the ice cream in the cone is h cm, the volume of ice cream is $\frac{7}{66} h^3$, using the approximate value $\pi = \frac{22}{7}$
- (ii) At what rate is the depth of ice cream in the cone decreasing when $h = 11$

Question 7.

(12 marks)

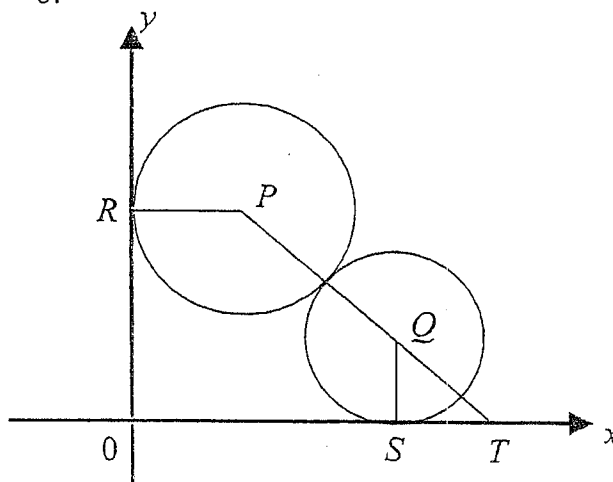
(Start on a new page)

- [4] (a) A particle is moving in a straight line. At time t seconds its velocity, v metres per second, and its displacement, x metres, are such that:

$$v^2 = 48 - 3x^2$$

- (i) Show that the motion is simple harmonic.
- (ii) Find the amplitude of the motion.
- (iii) Determine the particle's maximum speed.
- (iv) Determine the particle's maximum acceleration.

- [8] (b) The diagram shows two touching circles, with centres P and Q . The circle with centre P has a radius of 4 units and touches the y -axis at R . The circle with centre Q has a radius of 3 units and touches the x -axis at S . PQ produced meets the x -axis at T and $\angle QTS = \theta$.



- (i) Show that $OR = 3 + 7 \sin \theta$ and $OS = 4 + 7 \cos \theta$
- (ii) Show that $RS^2 = 42 \sin \theta + 56 \cos \theta + 74$
- (iii) Hence express RS^2 in the form $74 + r \cos(\theta - \alpha)$, clearly stating the values of r and α .
- (iv) Find the maximum length of RS and the value of θ for which this occurs.

END OF EXAMINATION

1. (a) i) $\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$

ii) $\tan 105^\circ = \tan(45^\circ + 60^\circ)$
 $= \frac{\tan 45 + \tan 60}{1 - \tan 45 \tan 60}$

$$= \frac{1 + \sqrt{3}}{1 - \sqrt{3}}$$

$$= \frac{1 + \sqrt{3}}{1 - \sqrt{3}} \times \frac{1 + \sqrt{3}}{1 + \sqrt{3}}$$

$$= \frac{-2 - \sqrt{3}}{1}$$

(b) $\frac{2x+1}{2x-1} \geq 2$

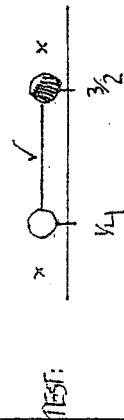
Critical Value when $2x-1=0$
 ie. $x = \frac{1}{2}$

Now, $2x+1 \geq 2(2x-1)$

$$2x+1 \geq 4x-2$$

$$3 \geq 2x$$

$$\therefore x \leq \frac{3}{2}$$



$$\therefore \frac{1}{2} < x \leq \frac{3}{2}$$

(c) $\int_0^{\pi/4} \cos x \cdot \sin^2 x \, dx = \left[\frac{1}{3} \sin^3 x \right]_0^{\pi/4}$

$$= \frac{1}{3} (\sin^3 \frac{\pi}{4} - \sin^3 0)$$

$$= \frac{1}{3} \left(\frac{1}{\sqrt{2}} \right)^3 - 0$$

$$= \frac{1}{6\sqrt{2}} = \frac{\sqrt{2}}{12} \approx 0.1178.$$

(d) $x^6 - 9x^3 + 8 = 0$

Let $a = x^3$

$$\therefore a^2 - 9a + 8 = 0$$

$$(a-8)(a-1) = 0$$

$$\therefore a = 8, 1.$$

Now, $x^3 = 8 \rightarrow x = 2$

$x^3 = 1 \rightarrow x = 1$

Al-

Aw-

Al-

Al-

Aw-

Aw-①

Al-1

Al-1

Aw-③

Al-1

Al-1

Aw-③

Q2. (a) $\int_0^1 x \log x - x = (x \cdot \frac{1}{2}x + \log x \cdot 1) - 1$

$= 1 + \log x - 1$

$= \log x$

(i) $\int_2^e \log x \, dx = [x \log x - x]_2^e$

$= (e \log e - e) - (2 \log 2 - 2)$

$= (e - e) - 2(\log 2 - 1)$

$= 2(1 - \log 2)$

(b) i) CF = CE (tangents from an external point are equal).

$\therefore \triangle ECF$ is isosceles.

$\therefore \angle EFC = \angle CEF = 65^\circ$ (base $\angle$'s isosceles)

(ii) $\angle EOF = \angle CEF = 65^\circ$ (alt. segment theorem)

(c) i) $\cos 2\alpha = 2 \cos^2 \alpha - 1$

let $\alpha = 3x$

$\therefore \cos 6x = 2 \cos^2 3x - 1$

(ii) $V = \pi \int y^2 \, dx$

$= \pi \int_0^{\frac{\pi}{6}} \cos^2 3x \, dx$

$= \pi \int_0^{\frac{\pi}{6}} \frac{1}{2} (\cos 6x + 1) \, dx$

$= \frac{\pi}{2} [x - \frac{1}{6} \sin 6x]_0^{\frac{\pi}{6}}$

$= \frac{\pi}{2} \left[\left(\frac{\pi}{6} - \frac{1}{6} \sin \pi \right) - (0 - \frac{1}{6} \sin 0) \right]$

Al-1

Aw-2

Al-1

Aw-2

Al-1

Aw-2

Aw-1

Aw-1

Al-1

Al-1

Al-1

↓

$= \frac{\pi}{2} \left(\frac{\pi}{6} \right)$

$= \frac{\pi^2}{12} \text{ units}^3$

Q3. (a) i) Let $f(x) = x^2 - 4 + \log_e x$

$\therefore f(1) = 1^2 - 4 + \log_e(1) = -3$

$f(2) = 2^2 - 4 + \log_e(2) = 0.693$

Since $f(x)$ changes sign, the root lies between 1 and 2.

(ii) Half the interval of 1 and 2 is $1\frac{1}{2}$.

$f(1\frac{1}{2}) = 1.5^2 - 4 + \log_e(1.5) = -1.34$

Since $f(1\frac{1}{2}) < 0$, the root lies between 1.5 and 2

Half the interval of 1.5 and 2 is 1.75

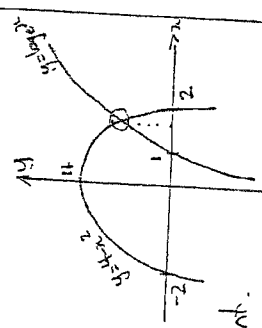
$f(1.75) = 1.75^2 - 4 + \log_e(1.75) = -0.378$

Since $f(1.75) < 0$, the root lies between 1.75 and 2

(iii) The root is the intersection of $y = \log_e x$ and $y = 4 - x^2$.

ie. $\log_e x = 4 - x^2$

There is only one such point.



Aw-

Q3. continued...

(b) $y = \frac{x^2 - 2x - 3}{x - 1}$

i) When $x = 0$: $y = \frac{-3}{-1} \rightarrow y = 3$

When $y = 0$: $x^2 - 2x - 3 = 0$
 $(x - 3)(x + 1) = 0 \rightarrow x = 3, -1$

ii) Vertical Asymptotes: when $x - 1 = 0$
 $\rightarrow x = 1$

Oblique Asymptotes: when $y = x - 1$

DIVIDE: $\frac{x^2 - 2x + 1}{x - 1} = \frac{4}{x - 1}$

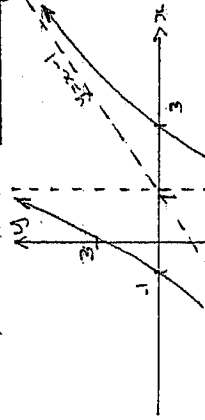
iii) $\frac{dy}{dx} = \frac{(x - 1)(2x - 2) - (x^2 - 2x - 3)}{(x - 1)^2}$
 $= \frac{2x^2 - 2x - 2x + 2 - x^2 + 2x + 3}{(x - 1)^2}$
 $= \frac{x^2 - 2x + 5}{(x - 1)^2}$

TP when $\frac{dy}{dx} = 0$

i.e. $x^2 - 2x + 5 = 0$

Since $\Delta < 0$, there are NO stationary points.

iv)



Q4.

(a) i) AP: $\beta - \alpha = \gamma - \beta$

$\therefore 2\beta = \alpha + \gamma$

ii) $\alpha + \beta + \gamma = \frac{36}{3} = 12$

iii) $(\alpha + \gamma) + \beta = 12$

$2\beta + \beta = 12$

$3\beta = 12 \rightarrow \beta = 4$

Now, $\alpha + \gamma = 12 - \beta = 8$ ——— ①

Also, $\alpha\beta\gamma = -\frac{d}{a} = -\frac{-24}{3} = 8$

$\therefore \alpha\gamma = \frac{8}{\beta} = 2$ ——— ②

Eq. ①: $\gamma = 8 - \alpha$

Sub into ②: $\alpha(8 - \alpha) = 2$

$8\alpha - \alpha^2 - 2 = 0$

$(2\alpha + 1)(\alpha - 7) = 0$

$\therefore \alpha = -\frac{1}{2}, 7$

When $\alpha = -\frac{1}{2}, \gamma = 7\frac{1}{2}$
 $\alpha = 7, \gamma = -\frac{1}{2}$

$\therefore$ The roots are $-\frac{1}{2}, \frac{3}{2}, \frac{7}{2}$

b) i) $\sum_{n=1}^n (5n - 4) = 1 + 6 + 11 + \dots + (5n - 4)$

ii) Prove: $1 + 6 + 11 + \dots + (5n - 4) = \frac{1}{2}n(5n - 3)$

* Prove true for $n = 1$:

$1 + 5 = 6 = \frac{1}{2} \cdot 1 \cdot (5 \cdot 1 - 3)$

Q4. continued---

* Assume true for $n=k$:

ie. $1+6+11+\dots+5k-4 = \frac{1}{2}k(5k-3)$

* Prove true for $n=k+1$:

RTP: $[1+6+11+\dots+5k-4] + 5(k+1) = \frac{1}{2}(k+1)(5(k+1)-3)$

ie. $1+6+11+\dots+(5k-4)+(5k+1) = \frac{1}{2}(k+1)(5k+2)$

LHS = $\frac{1}{2}k(5k-3) + (5k+1)$

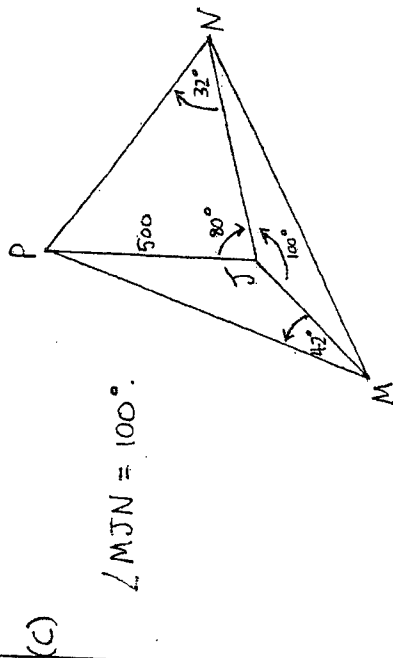
= $\frac{1}{2}[5k^2 - 3k + 10k + 2]$

= $\frac{1}{2}[5k^2 + 7k + 2]$

= $\frac{1}{2}(k+1)(5k+2)$

= RHS.

Since proved true for $n=k+1$ and show true for $n=1$; then the result must be true for $n=2, 3, \dots$ by the process of Mathematical Induction.



Q4 continued---

i) In $\triangle JMP$: $\tan 42^\circ = \frac{500}{JM}$

$\therefore JM = \frac{500}{\tan 42^\circ}$

In $\triangle JNP$: $\tan 32^\circ = \frac{500}{JN}$

$\therefore JN = \frac{500}{\tan 32^\circ}$

ii) In $\triangle JMN$: By the Cosine Rule,

$MN^2 = \left(\frac{500}{\tan 42^\circ}\right)^2 + \left(\frac{500}{\tan 32^\circ}\right)^2 - 2\left(\frac{500}{\tan 42^\circ}\right)\left(\frac{500}{\tan 32^\circ}\right)\cos 100^\circ$

$\therefore MN = 1050.2 \text{ metres}$

Q5. (a) i) $M_{Pa} = \frac{ap^2 - aq^2}{2ap - 2aq} = \frac{a(p-q)(p+q)}{2a(p-q)} = \frac{p+q}{2}$

PQ parallel to $y=x$ }
ie. $m=1$. } $\therefore \frac{p+q}{2} = 1$

$p+q=2$

ii) $m = -\frac{1}{p}$ at $P(2ap, ap^2)$

$y-y_1 = m(x-x_1)$

$\therefore y - ap^2 = -\frac{1}{p}(x - 2ap)$

$yp - ap^3 = -x + 2ap$

$\therefore x + py = 2ap^2 + ap^3$ ———— (1)

iii) Similarly, eqn. of normal at Q is:

$x + qy = 2aq^2 + aq^3$ ———— (2)

Eq. (1)-(2): $y(p-q) = 2a(p-q) + a(p^3 - q^3)$

$y = 2a + a(p^2 + q^2 + pq)$

Al-

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Aw

Aw-

Al-

Q3. Continued...

Sub into Eq. ①: $x + p(2a + pq^2 + pq^2 + pq^2) = 2ap + ap^3$

$$\therefore x = -2ap - ap^3 - apq^2 - apq^2 - apq^2 + 2ap + ap^3$$

$$x = -apq(pq)$$

But $pq = 2 \Rightarrow x = -2apq$

$\therefore$ Coordinates of R: $x = -2apq$ ——— ③
 $y = 2a + a(p^2 + q^2 + pq) =$ ④

④ Eq. ②: $y = 2a + a[(pq)^2 - pq]$

$$y = 2a + a(4 - pq)$$

$$y = 6a - apq$$

Eq. ③: $pq = \frac{-x}{2a}$

$$\therefore y = 6a + \frac{x}{2}$$

$$2y = 12a + x$$

$$\therefore x - 2y + 12a = 0$$

is the eqn of the locus of R.

(b) ① $T = T_0 + Ae^{kt}$

$$\frac{dT}{dt} = 0 + Ae^{kt} \times k$$

$$= (T - T_0) \times k$$

$$\therefore \frac{dT}{dt} = k(T - T_0)$$

Q5. Continued...

② $T_0 = -40^\circ$, $T = 19^\circ$

(a) When $t = 0$: $24 = -40 + Ae^0$
 $\therefore A = 64$

(b) When $t = 5$: $19 = -40 + 64e^{5k}$
 $59 = 64e^{5k}$

$$e^{5k} = \frac{59}{64}$$

$$5k = \ln\left(\frac{59}{64}\right)$$

$$\therefore k = \frac{1}{5} \ln\left(\frac{59}{64}\right)$$

ie. $k = -0.0163$ (3sf)

8) For $T = 0$:

$$0 = -40 + 64e^{-0.0163t}$$

$$e^{-0.0163t} = \frac{40}{64}$$

$$-0.0163t = \ln\left(\frac{40}{64}\right)$$

$$\therefore t = \frac{\ln\left(\frac{40}{64}\right)}{-0.0163}$$

ie. $t = 28.9$ seconds

Aw-②

Aw-③

Aw-①

A

Aw

(a) $\tan 2x = 1$

(b) $\dot{x} = 0$

$$\vec{r}_L = V \cos \theta$$

$$\begin{aligned} \vec{r} &= V \cos \theta & \dot{r} &= -V \sin \theta \\ \dot{\theta} &= V \sin \theta & \ddot{\theta} &= -V \cos \theta \end{aligned}$$

$$dy = -5t^2 + v^4 \sin \theta$$

① At $t = 1.5$, $x = 60$:

$$\theta_{sq}(5.1) \lambda = 0.9$$

$$V \cos \theta = 40 - 0$$

$$At + 1.5, y = 2.25:$$

$$2 \cdot 25 = -5(1.5)^2 + V(1.5) \sin \theta$$

$$V \sin \theta = a \quad \text{--- (2)}$$

Eq. ② ÷ ①: $\tan \theta = \frac{9}{40}$ $\rightarrow$ $\theta = 12.68^\circ$

Sub into Eq. (1): $V \cos(12.68^\circ) = 40 \rightarrow$

ii) When $y=0$: $-5t^2 + 4\sin\theta = 0$

$$+(-5t + v \sin \theta) = 0$$

$$\therefore f = 0 \text{ or } f = \frac{V \sin \theta}{5}$$

i.e. $t = 0, \frac{9}{5}$.

When $t = \frac{9}{5} :$

$$x = \sqrt{t} \cos \theta$$

$$x = 41 + \frac{9}{5} + \cos 12.68^\circ$$

$$\therefore x = 72 \text{ metres.}$$

Aw-2

Aw-2

Continued...

$$\frac{22}{7} = \frac{h}{22}$$

$$22r = 7h \Rightarrow r = \frac{7h}{22}$$

$$\begin{aligned} V &= \frac{1}{3} \pi r^2 h \\ &= \frac{1}{3} \cdot \frac{22}{7} \cdot \left(\frac{70}{22} \right)^2 \cdot h \\ &= \frac{22}{21} \cdot \frac{49h^2}{484} \cdot h \end{aligned}$$

$$\therefore \sqrt{\frac{1078h^3}{10164}} = \left(\frac{7}{66}h^3\right)$$

③ Now, $\frac{dV}{dh} = \frac{7}{22} h^2$ and $\frac{dV}{dt} = 7 \text{ cm/min.}$

$$\therefore \frac{dp}{dh} = \frac{dp}{dV} \cdot \frac{dV}{dh}$$

$$= 7. \frac{22}{7} \pi^2$$

$$\therefore \frac{dy}{dt} = \frac{22}{5}$$

When $h=11$, $\frac{dy}{dt} = \frac{22}{121} = \frac{2}{11} \text{ cm/min}$

A1-1

Aw-C

A-1-1

Aw-2

Q7.

(a) (i) $\frac{1}{2} v^2 = 24 - \frac{3}{2} x^2$

$\frac{d}{dx}(\frac{1}{2} v^2) = -3x$
 $\therefore \frac{dv}{dt} = -3x$

Since the acceleration is proportional to its distance from the origin, it is moving in SHM.

(ii) $v^2 = 3(16 - x^2)$

ie. $v^2 = 3(16 - x^2) \rightarrow \therefore x = 4 \text{ metres}$

(iii) Max. Speed when $x=0$:

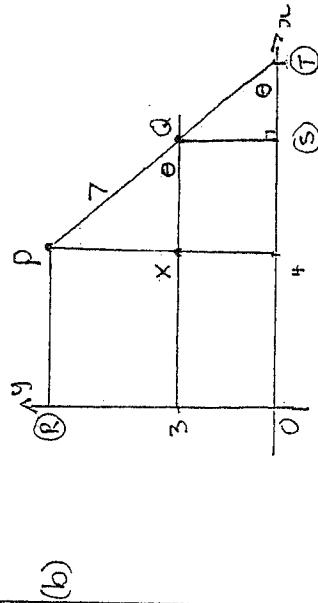
ie. $v^2 = 3(16-0)$

$\therefore v = \sqrt{48} \text{ ms}^{-1}$ is the max. speed.

(iv) Max. Acceleration when $x = \pm 4$

ie. $\frac{dv}{dt} = -3x = -4$

$\therefore \frac{dv}{dt} = 12 \text{ ms}^{-2}$ is the max. acceleration



(i) $OR = 3 + PX = 3 + 7 \sin \theta$
 $OS = 4 + QX = 4 + 7 \cos \theta$

Aw-①

Q7.

Continued...

(ii) $RS^2 = OR^2 + OS^2$ (Pythagoras' Theorem)
 $= (3 + 7 \sin \theta)^2 + (4 + 7 \cos \theta)^2$
 $= 9 + 49 \sin^2 \theta + 49 \sin^2 \theta + 16 + 56 \cos \theta + 49 \sin^2 \theta$
 $= 74 + 42 \sin \theta + 56 \cos \theta$

Aw

(iii) Let $42 \sin \theta + 56 \cos \theta = r \cos(\theta - \alpha)$

where $r = \sqrt{42^2 + 56^2} = 70$

$\alpha = \tan^{-1}(\frac{42}{56}) = 36^\circ 52'$

Al-

$\therefore 42 \sin \theta + 56 \cos \theta = 70 \cos(\theta - 36^\circ 52')$

$\therefore RS^2 = 74 + 70 \cos(\theta - 36^\circ 52')$

Aw-

(iv) Maximum value of RS occurs when:

$\cos(\theta - 36^\circ 52') = 1$

ie. when $\theta = 36^\circ 52'$.

Al-

$\therefore RS^2 = 74 + 70 = 144$

ie. $RS = 12 \text{ units}$

Aw-